

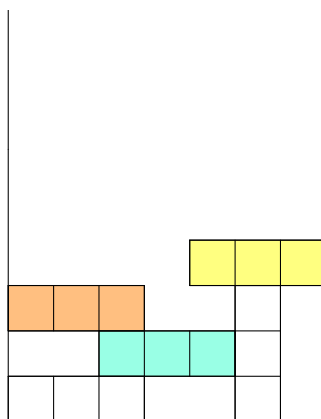
Problem 77. Alex is playing a computer game with falling two-dimensional blocks. The playing field is 7 units wide and infinitely tall with a bottom border. Initially the entire field is empty. Each turn, the computer gives Alex a 1×3 solid rectangular piece of three unit squares. Alex must decide whether to orient the piece horizontally or vertically and which column(s) the piece should occupy (3 consecutive columns for horizontal pieces, 1 column for vertical pieces). Once he confirms his choice, the piece is dropped straight down into the playing field in the selected columns, stopping all three of the piece's squares as soon as the piece hits either the bottom of the playing field or any square from another piece. All of the pieces must be contained completely inside the playing field after dropping and cannot partially occupy columns.

If at any time a row of 7 spaces is all filled with squares, Alex scores a point.

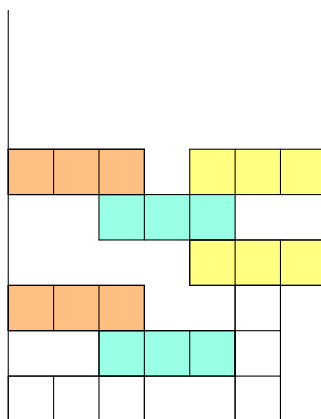
Unfortunately, Alex is playing in invisible mode, which prevents him from seeing the state of the playing field or how many points he has, and he already arbitrarily dropped some number of pieces without remembering what he did with them or how many there were.

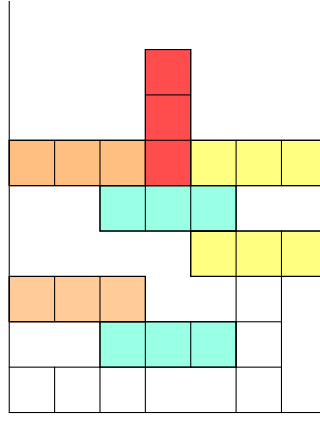
Find a strategy for which Alex can correctly announce "I have earned at least one more point" and know that he is correct.

Solution. Drop a horizontal piece occupying columns 3, 4, and 5. Then drop two horizontal pieces occupying columns 1, 2, and 3 and columns 5, 6, and 7. We illustrate this in the diagram below (the new pieces dropped at this step are in gray).



Since the three pieces we just dropped span every column, one of latter two pieces is guaranteed to be our highest piece on the board at this point. Without loss of generality, we assume the piece on the right is now our highest piece (as in the diagram above). We repeat this pattern. When we drop the horizontal piece occupying columns 3, 4, and 5, it will be stopped by the horizontal piece occupying columns 5, 6, and 7 that we dropped previously. Each of our successive pieces will then be stopped by this piece. Thus, our board looks like this:





At this point, Alex can simply drop one vertical piece in the middle and exclaim “I have earned at least one more point,” as desired. $\square$

Remark. If Alex repeatedly dropped vertical pieces in each column, he would eventually earn a point, but he cannot know when that has happened.

Problem 78. For $n > 1$, let a_n be the number of zeroes that $n!$ ends with when written in base n . Find the maximum value of $\frac{a_n}{n}$.

Solution. The maximum value of $\frac{a_n}{n}$ is $\frac{1}{2}$.

Now observe that the base 2 representation of $2!$ is 10, which ends with 1 zero. So $a_2 = 1$, and $\frac{a_2}{2} = \frac{1}{2}$. So it suffices to show that $\frac{a_n}{n}$ is at most $\frac{1}{2}$ for all positive integers n .

Notice that the number of zeroes that $n!$ ends with when written in base n is the same as the largest power of n that divides $n!$.

Fix n and write $n = p^k m$, where p is some prime dividing n and m is relatively prime to p . Then a_n is at most the largest power of p^k dividing $n!$. By Legendre’s formula, the largest power of p dividing $n!$ (usually denoted by $v_p(n!)$) is given by

$$v_p(n!) = \sum_{j=1}^{\infty} \left\lfloor \frac{n}{p^j} \right\rfloor.$$

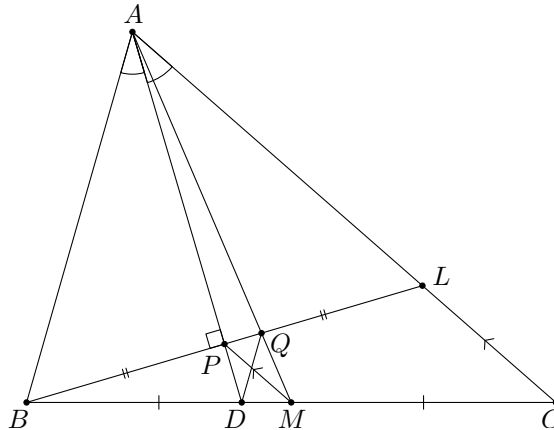
So we have

$$\frac{a_n}{n} \leq \frac{1}{nk} \sum_{j=1}^{\infty} \left\lfloor \frac{n}{p^j} \right\rfloor < \frac{1}{nk} \sum_{j=1}^{\infty} \frac{n}{p^j} = \frac{1}{k} \sum_{j=1}^{\infty} \frac{1}{p^j} = \frac{1}{k(p-1)}.$$

Therefore if n has any prime divisor greater than 2, or is divisible by the square of any prime, we have $\frac{a_n}{n} < \frac{1}{2}$. $\square$

Problem 79. Let $\triangle ABC$ be a triangle with $AB < AC$. Let the angle bisector of $\angle BAC$ meet BC at D , and let M be the midpoint of BC . Let P be the foot of the perpendicular from B to AD . Extend BP to meet AC at Q . Show that DQ is parallel to AB .

Solution 1.



Let BP meet AC at L . By construction, $\angle APB = \angle APL = 90^\circ$ and $\angle PAL = \angle PAB$ (because AP is the angle bisector from A), and since they share side AP , and so $\triangle APB \cong \triangle APL$. Consequently P is the midpoint of BL . Since M is the midpoint of BC , by the Midpoint Theorem, we have $PM \parallel AC$. Hence $\triangle DPM \sim \triangle DAC$ and so

$$\frac{MP}{AC} = \frac{DM}{DC}.$$

To finish, it enough to show that $\triangle MDQ \sim \triangle MBA$, which means that it suffices to show that

$$\frac{MD}{DB} = \frac{MQ}{QA}.$$

Since $\angle AQL = \angle PQM$ and $PM \parallel AL$, we see that $\triangle QPM \sim \triangle QLA$ and so

$$\frac{MQ}{QA} = \frac{MP}{AL}$$

Hence we just need to show that

$$\frac{MD}{DB} = \frac{MP}{AL}.$$

Notice that if $\frac{a}{b} = \frac{c}{d} \neq \frac{1}{2}$, then $\frac{a}{b-2a} = \frac{c}{d-2c}$. Applying this to

$$\frac{MP}{AC} = \frac{DM}{DC}$$

gives us

$$\frac{MP}{AC - 2PM} = \frac{MD}{CD - 2MD}$$

But $2PM = LC$ (by the Midpoint Theorem), and so

$$\frac{MP}{AC - 2PM} = \frac{MP}{AC - LC} = \frac{MP}{AL}$$

Similarly, we have $CM = BD + DM$, so $CD = BD + DM + DM = BD + 2DM$. Thus,

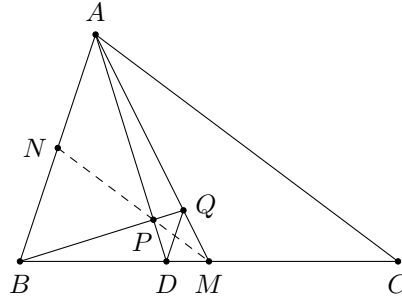
$$\frac{MD}{CD - 2MD} = \frac{MD}{BD}$$

Putting these together, we have

$$\frac{MP}{AL} = \frac{MD}{BD}$$

as desired. □

Solution 2.



Let N be the midpoint of AB . As $\angle APB = 90^\circ$, we have $NP = NA = NB$. Therefore, $\angle BNP = 2\angle BAP = \angle BAC = \angle BNM$. This implies N, P, M are collinear. By Ceva's theorem, we obtain $\frac{BD}{DM} \times \frac{MQ}{QA} \times \frac{AN}{NB} = 1$. This yields $\frac{BD}{DM} = \frac{AQ}{QM}$. It follows that AB and QD are parallel. □

Problem 80. Let $a_1, a_2, \dots, a_{2020}$ be a sequence of integers. Initially, $a_1 = 1, a_2 = -1$ and the remaining numbers are 0. After every second, we perform the following process on the sequence: for $k = 1, 2, \dots, 2019$, replace a_k with $a_k + a_{k+1}$, and replace a_{2020} with $a_{2020} + a_1$. (All of this is done simultaneously, so each new term is the sum of two terms of the sequence from before any replacements.) Prove that for any integer M , there is some index k and some time t for which $|a_k| > M$ at time t .

Solution 1. Let $N = 2020$ and fix a positive integer M . Throughout this solution we consider indices modulo N . Let $a_{n,t}$ denote a_n after t seconds, with a_n being the original sequence, i.e. $a_n = a_{n,0}$. We want to show that for some positive integers k and t we have $|a_{k,t}| > M$.

Now observe that

$$a_{1,1} = a_1 + a_2,$$

$$a_{1,2} = a_{1,1} + a_{2,1} = (a_1 + a_2) + (a_2 + a_3) = a_1 + 2a_2 + a_3,$$

$$a_{1,3} = a_{1,2} + a_{2,2} = (a_1 + 2a_2 + a_3) + (a_2 + 2a_3 + a_4) = a_1 + 3a_2 + 3a_3 + a_4.$$

Continuing this pattern, we claim that

$$a_{n,t} = a_n + \binom{t}{1}a_{n+1} + \binom{t}{2}a_{n+2} + \dots$$

for all positive integers n . We show this by induction on t . For $t = 0$, the claim is true by definition. Suppose the claim holds for $t - 1$ and we will show it holds for t . By induction assumption, we have

$$\begin{aligned} a_{n,t} &= a_{n,t-1} + a_{n+1,t-1} \\ &= \left(a_n + \binom{t-1}{1}a_{n+1} + \binom{t-1}{2}a_{n+2} + \dots \right) + \left(a_{n+1} + \binom{t-1}{1}a_{n+2} + \binom{t-1}{2}a_{n+3} + \dots \right) \\ &= a_n + \left(\binom{t-1}{1} + 1 \right) a_{n+1} + \left(\binom{t-1}{2} + \binom{t-1}{1} \right) a_{n+2} + \dots \\ &= a_n + \binom{t}{1}a_{n+1} + \binom{t}{2}a_{n+2} + \dots, \end{aligned}$$

which proves the claim.

Now observe that most of these terms are actually 0. Getting rid of all the 0's and noting that $a_1 = 1, a_2 = -1$, we have

$$\begin{aligned} a_{r,t} &= a_r + \left(\binom{t}{N+1-r}a_{N+1} + \binom{t}{2N+1-r}a_{2N+1} + \dots \right) + \left(\binom{t}{N+2-r}a_{N+2} + \binom{t}{2N+2-r}a_{2N+2} + \dots \right) \\ &= \sum_{s \equiv 1-r \pmod{N}} \binom{t}{s} - \sum_{s \equiv 2-r \pmod{N}} \binom{t}{s}. \end{aligned}$$

Assume on the contrary that for all positive integers n, t , $|a_{n,t}| < M$. Now let $\omega = \cos \frac{2\pi}{N} + i \sin \frac{2\pi}{N}$. Then, by the triangle inequality,

$$\left| \sum_{r=1}^N a_{r,t} \omega^r \right| \leq \sum_{r=1}^N |a_{r,t}| \cdot 1 < NM. \quad (*)$$

However, note that

$$\begin{aligned} \sum_{r=1}^N a_{r,t} \omega^r &= \sum_{r=1}^N \left(\sum_{s \equiv 1-r \pmod{N}} \binom{t}{s} \omega^r - \sum_{s \equiv 2-r \pmod{N}} \binom{t}{s} \omega^r \right) \\ &= \omega \left(\sum_{s=1}^t \binom{t}{s} \omega^{-s} - \sum_{s=1}^t \binom{t}{s} \omega^{-s+1} \right) \\ &= \omega \left((1 + \omega^{-1})^t - \omega (1 + \omega^{-1})^t \right) \\ &= (\omega - \omega^2) (1 + \omega^{-1})^t. \end{aligned}$$

Taking magnitudes, we have

$$\left| \sum_{r=1}^N a_{r,t} \omega^r \right| = |\omega - \omega^2| \cdot |1 + \omega^{-1}|^t.$$

Note that

$$|1 + \omega^{-1}| = \left| 1 + \cos \frac{2\pi}{N} - i \sin \frac{2\pi}{N} \right| = \sqrt{2 \left(1 + \cos \frac{2\pi}{N} \right)} > 1.$$

Hence we can make $|1 + \omega^{-1}|^t$ arbitrarily large by considering a sufficiently large t and this contradicts (*). Therefore for some positive integers k and t we have $|a_{k,t}| > M$. $\square$

Solution 2.

Let $a_{n,t}$ denote a_n after t seconds, with a_n being the original sequence, i.e. $a_n = a_{n,0}$. The index n is taken modulo $N = 2020$.

Suppose on the contrary that $|a_{n,t}|$ is bounded. Let M be the maximum value of $a_{n,t}$ and let m be the minimum value of $a_{n,t}$. Suppose $a_{n_1,t_1} = M$ and $a_{n_2,t_2} = m$. Note that

$$\begin{aligned} a_{n,t} &= a_{n,t-1} + a_{n+1,t-1} \\ &= a_{n,t-2} + 2a_{n+1,t-2} + a_{n+2,t-2} \\ &= a_{n,t-3} + 3a_{n+1,t-3} + 3a_{n+2,t-3} + a_{n+3,t-3} \\ &= a_{n,t-3} + 3a_{n+1,t-2} + a_{n+3,t-3}. \end{aligned} \quad (1)$$

Taking $n = n_1 - 1$ and $t = t_1 + 2$ in (1), we obtain

$$M \geq a_{n_1-1, t_1+2} = a_{n_1-1, t_1-1} + 3a_{n_1, t_1} + a_{n_1+2, t_1-1} \geq m + 3M + m,$$

which implies $-M \geq m$. Taking $n = n_2 - 1$ and $t = t_2 + 2$ in (1), we obtain

$$m \leq a_{n_2-1, t_2+2} = a_{n_2-1, t_2-1} + 3a_{n_2, t_2} + a_{n_2+2, t_2-1} \leq M + 3m + M,$$

which implies $-M \leq m$. Combining the two results, we find that $-M = m$. Therefore, all the above inequalities are equalities. In particular, we have $a_{n_1-1, t_1-1} = m$ and $a_{n_2-1, t_2-1} = M$.

From above, whenever $a_{n_1, t_1} = M$, we have $a_{n_1-1, t_1-1} = m$. Then we can take $n_2 = n_1 - 1$ and $t_2 = t_1 - 1$ to get $a_{n_1-2, t_1-2} = M$. Inductively, we find that $M = |a_{n_1, t_1}| = |a_{n_1-1, t_1-1}| = |a_{n_1-2, t_1-2}| = \dots = |a_{n_1-t_1, 0}| \leq 1$. This means $|a_{n, t}| \leq M \leq 1$ for any n and t . Clearly, this is impossible. For example, one finds directly that $a_{1,3} = -2$. This is a contradiction, which means $|a_{n, t}|$ is unbounded. □

Solution 3.

Let $a_{n, t}$ denote a_n after t seconds, with a_n being the original sequence, i.e. $a_n = a_{n, 0}$. The index n is taken modulo 2020. Suppose on the contrary that each $a_{n, t}$ is bounded. Then the sequence $(a_{1, t}, a_{2, t}, \dots, a_{2020, t})$ only has a finite number of possibilities. Therefore, this sequence must repeat at some stage, and is eventually periodic.

Choose the smallest $t_0 \geq 1$ and $d \geq 1$ such that $a_{n, t+d} = a_{n, t}$ for all $1 \leq n \leq 2020$ and $t \geq t_0$. We claim that $t_0 = 1$. To prove this claim, we first observe the following facts:

- $a_{1, t} + a_{2, t} + \dots + a_{2020, t} = 0$ for any $t \geq 0$
- $a_{1, t} + a_{3, t} + \dots + a_{2019, t} = a_{2, t} + a_{4, t} + \dots + a_{2020, t} = 0$ for any $t \geq 1$

These can be easily proved by induction. Now, suppose $t_0 > 1$. Note that $a_{n, t_0} = a_{n, t_0-1} + a_{n+1, t_0-1}$ and $a_{n, t_0+d} = a_{n, t_0+d-1} + a_{n+1, t_0+d-1}$ for every $1 \leq n \leq 2020$. If we let $a_{1, t_0-1} = a_{1, t_0+d-1} + k$ for some k , then

$$a_{2, t_0-1} = a_{2, t_0+d-1} - k, \quad a_{3, t_0-1} = a_{3, t_0+d-1} + k, \quad a_{4, t_0-1} = a_{4, t_0+d-1} - k, \dots, a_{2020, t_0-1} = a_{2020, t_0+d-1} - k.$$

But since $a_{1, t_0-1} + a_{3, t_0-1} + \dots + a_{2019, t_0-1} = a_{1, t_0+d-1} + a_{3, t_0+d-1} + \dots + a_{2019, t_0+d-1} = 0$, we must have $k = 0$. This shows the sequences $(a_{1, t_0-1}, a_{2, t_0-1}, \dots, a_{2020, t_0-1})$ and $(a_{1, t_0+d-1}, a_{2, t_0+d-1}, \dots, a_{2020, t_0+d-1})$ are the same, which contradicts our choice of t_0 . Therefore, we must have $t_0 = 1$. It follows that $(a_{1, d+1}, a_{2, d+1}, \dots, a_{2020, d+1}) = (0, -1, 0, 0, \dots, 0, 1)$. But then the solutions to the system

$$\left\{ \begin{array}{l} a_{1, d} + a_{2, d} = 0, \\ a_{2, d} + a_{3, d} = -1, \\ a_{3, d} + a_{4, d} = 0, \\ a_{4, d} + a_{5, d} = 0, \\ \dots \\ a_{2019, d} + a_{2020, d} = 0, \\ a_{2020, d} + a_{1, d} = 1 \end{array} \right.$$

are $(a_{1, d}, a_{2, d}, \dots, a_{2020, d}) = (m, -m, m-1, 1-m, m-1, 1-m, \dots, m-1, 1-m)$, which does not satisfy $a_{1, d} + a_{3, d} + \dots + a_{2019, d} = 0$. This is a contradiction. Therefore, $a_{n, t}$ is unbounded. □